

Diagram illustrating the geometry of a curved path (Bahn) and a local horizon (lokaler Horizont). A red vector $\mathbf{r}$ points from a red dot to a point on the path. A green vector $\mathbf{v}$ points vertically upwards from the same point. A blue arc indicates the angle γ between the tangent to the path and the vertical vector $\mathbf{v}$. A dashed line represents the local horizon.

The diagram shows a coordinate system with a horizontal x -axis and a vertical y -axis. A blue vector $\mathbf{e}_x$ points along the x -axis, and a blue vector $\mathbf{e}_y$ points along the y -axis. A red vector $\mathbf{r}$ originates from the origin and points to a point on a curved blue line labeled "Bahn" (trajectory). The angle between $\mathbf{r}$ and the x -axis is labeled ϑ . A green vector $d\mathbf{r}$ is tangent to the "Bahn" at that point. A red vector $\mathbf{r} + d\mathbf{r}$ originates from the tip of $\mathbf{r}$ and points to a point further along the "Bahn". A green vector $r d\vartheta$ is perpendicular to $\mathbf{r}$ and points in the direction of the "Bahn". A blue arc indicates the angle γ between $d\mathbf{r}$ and $r d\vartheta$. A dashed black line labeled "lokaler Horizont" (local horizon) is perpendicular to the "Bahn" at the point where $d\mathbf{r}$ is defined. A horizontal dashed line extends from the y -axis to the "Bahn".